Class 7 Biology

Complete Study Notes

Chapter 1: Nutrition in Plants

Chapter 2: Respiration in Organisms

NCERT | Science | Class VII

Includes: Key Points • Diagrams • Definitions • Q&A

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■ **KEY POINTS**

- ✥ Nutrition is the process of taking in food and using it for energy, growth, and repair.
- ✥ Plants are autotrophs — they make their own food through photosynthesis.
- ✥ Photosynthesis equation: 6CO■ + 6H■O + Sunlight → C■H■■O■ + 6O■
- ✥ Chlorophyll is the green pigment in leaves that traps solar energy.
- ✥ Stomata are tiny pores on leaves that allow CO■ in and O■ out.
- ✥ Roots absorb water and minerals from the soil.
- ✥ Some plants are heterotrophs — they depend on others for food.
- ✥ Parasitic plants (Cuscuta) take food from host plants.
- ✥ Insectivorous plants (Venus Flytrap, Pitcher Plant) trap insects for nitrogen.
- ✥ Symbiotic plants (Lichens, Rhizobium) involve mutual benefit.
- ✥ Saprotrophic organisms (Fungi, Bacteria) feed on dead/decaying matter.
- ✥ Rhizobium bacteria in legume roots fix atmospheric nitrogen into the soil.
- ✥ Saprophytes help in decomposition and recycling of nutrients.

1.1 What is Nutrition?

Nutrition is the process by which living organisms obtain food and use it to carry out life processes such as growth, repair, energy production, and maintenance of body functions. Without nutrition, no organism can survive.

There are two broad categories of nutrition:

- **Autotrophic Nutrition** — Organisms make their own food using simple substances (e.g., green plants).
- **Heterotrophic Nutrition** — Organisms depend on other organisms for food (e.g., animals, fungi, some plants).

Why do Plants Need Nutrition?

- To perform photosynthesis and produce food.
- To grow — produce new cells, leaves, roots, flowers, fruits.
- To repair damaged parts.
- To carry out all metabolic functions.

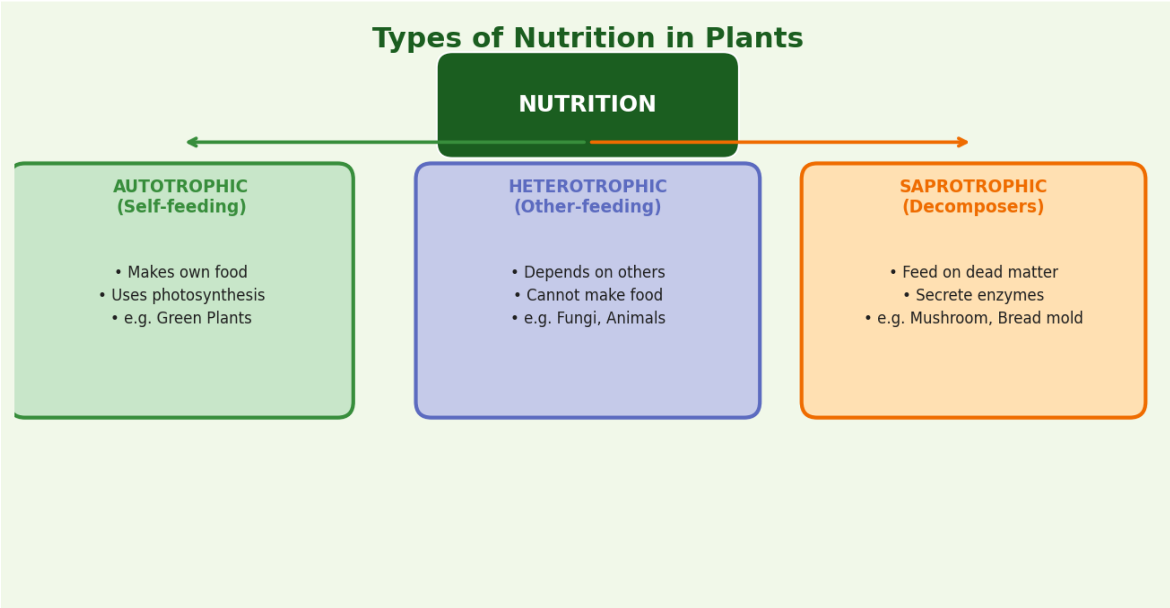

Fig 1.1 — Types of Nutrition in Plants

1.2 Autotrophic Nutrition & Photosynthesis

Photosynthesis is the process by which green plants prepare their own food using **sunlight**, **carbon dioxide (CO_2)** and **water (H_2O)** in the presence of the green pigment **chlorophyll** present in the chloroplasts of leaves.

■ KEY POINTS

❖ The word "Photosynthesis" comes from: Photo (light) + Synthesis (to make).

❖ It occurs mainly in leaves (and sometimes in green stems).

❖ The food produced is Glucose — a simple sugar.

❖ Oxygen is released as a by-product — vital for all animals.

❖ Chlorophyll is found in organelles called Chloroplasts.

Raw Materials for Photosynthesis

- **Carbon dioxide (CO_2)** — absorbed from air through stomata.
- **Water (H_2O)** — absorbed by roots from soil, transported via stem (xylem).
- **Sunlight** — solar energy that powers the reaction.
- **Chlorophyll** — the pigment that absorbs light energy.

Products of Photosynthesis

- **Glucose ($C_6H_{12}O_6$)** — used as food by the plant.

• **Oxygen (O■)** — released into the air through stomata.

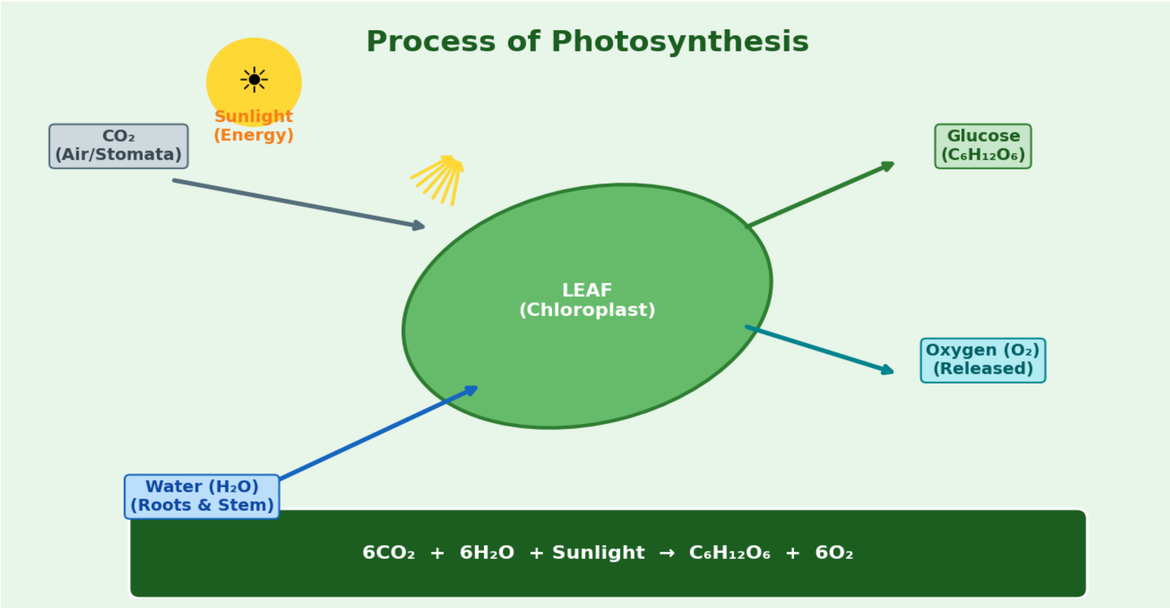

Fig 1.2 — Detailed Process of Photosynthesis in a Leaf

Chemical Equation of Photosynthesis

6CO■ + 6H■O	■■■■■Sunlight + Chlorophyll■■■■■	C■H■■O■ + 6O■

Role of Sunlight

Sunlight provides the energy needed to split water molecules (a process called **photolysis**). This energy is used to convert CO■ and H■O into glucose. Without sunlight, photosynthesis stops completely.

Role of Chlorophyll

Chlorophyll is the green-coloured pigment in plant cells. It is found inside **chloroplasts** (organelles in leaf cells). Chlorophyll absorbs the red and blue wavelengths of sunlight and reflects green light (which is why leaves look green). It converts the absorbed light energy into chemical energy.

■ Experiment to show that chlorophyll is necessary for photosynthesis: Variegated leaves (with green and white parts) — only the green parts with chlorophyll turn blue-black with iodine, proving starch (food) is produced only where chlorophyll is present.

1.3 Stomata and Gas Exchange

Stomata (singular: stoma) are microscopic pores (openings) found mainly on the lower surface of leaves. They are surrounded by two bean-shaped cells called **Guard Cells** which control their opening and closing.

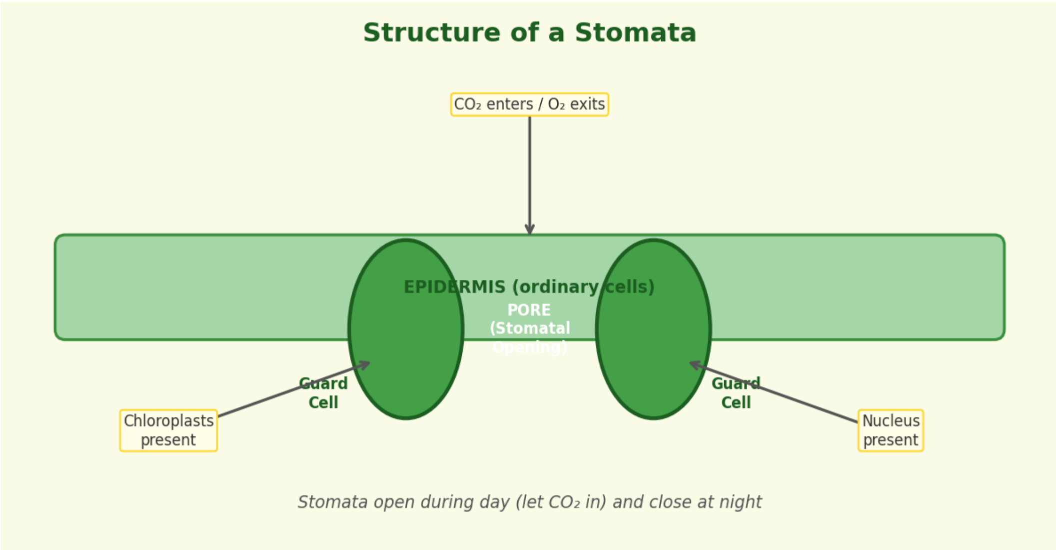

Fig 1.3 — Structure of Stomata and Guard Cells

Functions of Stomata

- **Gas Exchange**: CO■ enters the leaf; O■ and water vapour exit.
- **Transpiration**: Loss of water vapour through stomata helps in upward movement of water in the plant.

How Guard Cells Work

Condition	Guard Cell State	Stomata
Daytime / Water available	Turgid (swollen)	OPEN
Night / Drought	Flaccid (limp)	CLOSED

1.4 Heterotrophic Nutrition in Plants

Some plants cannot produce their own food. They depend on other organisms — these are called **Heterotrophic Plants**. Heterotrophic nutrition in plants can be of several types:

Parasitic	Obtains food from a living host plant (e.g., Cuscuta).
Insectivorous	Traps and digests insects to obtain nitrogen (e.g., Pitcher Plant, Venus Flytrap).
Symbiotic	Both organisms live together and benefit each other (e.g., Lichen, Rhizobium).

Saprotrophic	Feeds on dead and decaying organic matter (e.g., Mushroom, Rhizopus).

1.5 Parasitic Plants — Cuscuta (Dodder)

Cuscuta (Amarbel) is a common example of a parasitic plant. It has no chlorophyll and cannot perform photosynthesis. It wraps around a host plant (like wheat or tomato) and penetrates its tissues with special structures called **haustoria** to absorb water, minerals, and food directly.

Characteristics of Parasitic Plants

- Have no or very little chlorophyll — cannot photosynthesize.
- Completely dependent on host for nutrition.
- Possess **haustoria** — root-like structures that penetrate the host.
- Can weaken or kill the host plant over time.
- Cuscuta is orange/yellow coloured with no leaves.

■ *Partial parasites: Some plants like Sandalwood (Chandan) have their own leaves and can photosynthesize but also steal water and minerals from the host via haustoria. These are called hemi-parasites.*

1.6 Insectivorous Plants

Insectivorous (Carnivorous) Plants grow in soil that is very poor in nitrogen. Since they cannot get enough nitrogen from soil, they have evolved to trap and digest insects and small animals to fulfill their nitrogen requirement. They still perform photosynthesis for their energy needs.

Examples and Their Trap Mechanisms

Plant	Trap Type	Mechanism
Venus Flytrap	Snap Trap	Leaves snap shut when insect touches trigger hairs.
Pitcher Plant (Nepenthes)	Pitfall Trap	Funnel-shaped leaf collects rainwater; insect falls in and drowns. Digestive enzymes break it down.
Sundew (Drosera)	Flypaper Trap	Sticky mucilage on leaf hairs traps insects.

Bladderwort (Utricularia)	Suction Trap	Aquatic plant; tiny bladders suck in water microorganisms.

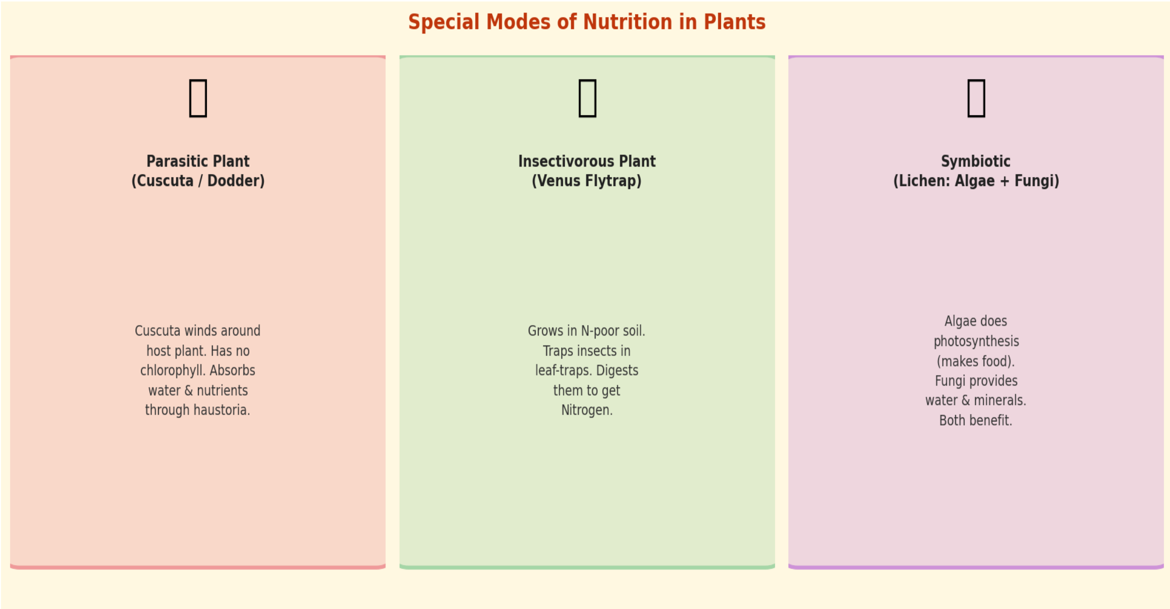

Fig 1.4 — Special Modes of Nutrition: Parasitic, Insectivorous, Symbiotic

1.7 Symbiotic Nutrition

Symbiosis means "living together" — two different organisms live in close association where both benefit from each other. This is called a mutually beneficial relationship.

Example 1: Lichen (Alga + Fungus)

- Lichen is a combination of an alga and a fungus.
- The alga performs photosynthesis and provides food (organic matter) to the fungus.
- The fungus provides shelter, water, and minerals to the alga.
- Both benefit — hence it is a mutualistic symbiosis.

Example 2: Rhizobium and Legumes

- **Rhizobium** is a type of bacteria that lives in the root nodules of leguminous plants like pea, gram, soybean, and beans.
- Rhizobium has the ability to **fix atmospheric nitrogen** — convert N■ gas from air into ammonia (NH■), which plants can use.
- The plant provides carbohydrates (food) to Rhizobium.
- Rhizobium provides nitrates/ammonia to the plant — essential for protein synthesis.
- Farmers use this naturally by planting legumes to enrich the soil with nitrogen.

1.8 Saprotrophic Nutrition

Saprotrophs are organisms that obtain nutrition by breaking down dead and decaying organic matter. They secrete digestive enzymes on the dead material, digest it externally (extracellular digestion), and then absorb the simpler molecules.

Examples of Saprotrophs

- Fungi: Mushroom, Rhizopus (bread mould), Aspergillus.
- Bacteria: Many decomposer bacteria in soil.

Importance of Saprotrophs

- Break down complex organic matter into simpler inorganic substances.
- Return nutrients to the soil — nitrogen, phosphorus, etc.
- Essential for recycling nutrients in the ecosystem.
- Help maintain soil fertility.
- Called "Nature's Decomposers" or "Nature's Recyclers."

1.9 Replenishment of Nutrients — Nitrogen Fixation

Plants take minerals from the soil during growth. If these minerals are not replenished, the soil becomes infertile. **Nitrogen** is especially important because it is required for making proteins and nucleic acids.

Nitrogen Fixation is the process of converting atmospheric nitrogen (N■) into compounds like ammonia or nitrates that plants can absorb.

How Nitrogen is Fixed

- **By Bacteria**: Rhizobium (in root nodules), Azotobacter, and Nostoc (blue-green algae) fix nitrogen naturally.
- **By Lightning**: During thunderstorms, lightning converts N■ + O■ → NO■■ (nitrates) which dissolve in rainwater and reach soil.
- **Industrial**: Haber Process: N■ + H■ → NH■ (Ammonia) — used to make fertilizers.
- **Returning Nutrients**: Dead plants/animals are decomposed by saprotrophs → nutrients return to soil.

1.10 Key Definitions — Chapter 1

Photosynthesis	Process by which green plants convert CO■ and H■O into glucose using sunlight and chlorophyll.
Chlorophyll	Green pigment in chloroplasts that absorbs sunlight for photosynthesis.
Stomata	Tiny pores on leaves for gas exchange (CO■ in, O■ out). Controlled by guard cells.
Chloroplast	Organelle in plant cells where photosynthesis occurs. Contains chlorophyll.
Autotroph	Organism that makes its own food (e.g., green plants).
Heterotroph	Organism that depends on others for food (e.g., animals, parasites).
Parasite	Organism that feeds on a living host, harming the host.
Haustoria	Special absorbing structures in parasitic plants that penetrate the host.
Symbiosis	Mutually beneficial relationship between two different organisms.
Saprotroph	Organism that feeds on dead/decaying organic matter (e.g., fungi).
Nitrogen Fixation	Conversion of atmospheric N■ into usable compounds like ammonia/nitrates.
Transpiration	Loss of water vapour through stomata of leaves.

1.11 Important Questions & Answers

Q	Answer
What is photosynthesis? Write its equation.	Photosynthesis is the process by which green plants make food using CO■, H■O, and sunlight in the presence of chlorophyll. Equation: 6CO■ + 6H■O + Sunlight → C■H■■O■ + 6O■
Where does photosynthesis occur?	Photosynthesis occurs in the chloroplasts, mainly in leaves. Chloroplasts contain the green pigment chlorophyll that absorbs sunlight.

What are stomata? What is their role?	Stomata are tiny pores on leaves surrounded by guard cells. They allow CO■ to enter and O■ and water vapour to exit. They open during the day and close at night.
What is the role of guard cells?	Guard cells are two bean-shaped cells surrounding each stoma. They control the opening and closing of stomata based on water availability and light.
What is a parasitic plant? Give an example.	A parasitic plant is one that obtains nutrients from a living host plant using haustoria. Example: Cuscuta (Amarbel / Dodder) — it has no chlorophyll and depends entirely on the host.
Why do insectivorous plants eat insects?	Insectivorous plants live in nitrogen-deficient soil. They trap and digest insects to obtain the nitrogen needed for making proteins.
What is Rhizobium? How does it help plants?	Rhizobium is a type of bacteria that lives in the root nodules of legume plants. It fixes atmospheric nitrogen into ammonia/nitrates which the plant absorbs for growth.
What is symbiosis? Give one example.	Symbiosis is a mutually beneficial relationship between two different organisms. Example: Lichen — algae provides food through photosynthesis; fungi provides water and shelter.
What is a saprotroph? Why are they important?	Saprotrophs are organisms that feed on dead/decaying matter. They are important because they decompose organic waste, return nutrients to the soil, and maintain ecosystem balance. e.g., Mushroom.
Differentiate between autotrophic and heterotrophic nutrition.	Autotrophic: organisms make their own food (e.g., green plants). Heterotrophic: organisms depend on others for food (e.g., animals, fungi, parasites).

■ KEY POINTS

- ✥ Respiration is the process by which organisms break down food (glucose) to release energy.
- ✥ Energy is released in the form of ATP (Adenosine Triphosphate).
- ✥ Respiration equation: C■H■■O■ + 6O■ → 6CO■ + 6H■O + Energy (ATP)
- ✥ Aerobic respiration requires oxygen — produces 38 ATP molecules.
- ✥ Anaerobic respiration does not require oxygen — produces only 2 ATP molecules.
- ✥ In muscles during intense exercise, lactic acid builds up causing cramps.
- ✥ Yeast performs anaerobic respiration: Glucose → Ethanol + CO■ (fermentation).
- ✥ Humans breathe using lungs; fish use gills; earthworms breathe through moist skin.
- ✥ Insects breathe through tiny holes called spiracles connected to tracheae.
- ✥ Breathing ≠ Respiration: Breathing is physical inhalation/exhalation; respiration is chemical.
- ✥ Mitochondria is called the "Powerhouse of the Cell" — site of aerobic respiration.
- ✥ All living organisms — plants, animals, microbes — perform respiration.

2.1 What is Respiration?

Respiration is the biological process by which living organisms **break down food** (mainly glucose) to release energy. This energy is used for all life processes — movement, growth, reproduction, and maintaining body temperature.

The energy is stored in the form of **ATP (Adenosine Triphosphate)** — often called the "energy currency" of cells. Respiration occurs in all living cells, 24 hours a day, 7 days a week. The site of respiration in the cell is the **Mitochondria**.

Difference: Breathing vs Respiration

Breathing	Respiration
Physical process	Chemical/Biochemical process
Involves movement of chest and diaphragm	Involves enzymatic reactions in cells
Brings O■ in; expels CO■ out	Breaks down glucose; releases energy

Occurs in respiratory organs	Occurs in all living cells

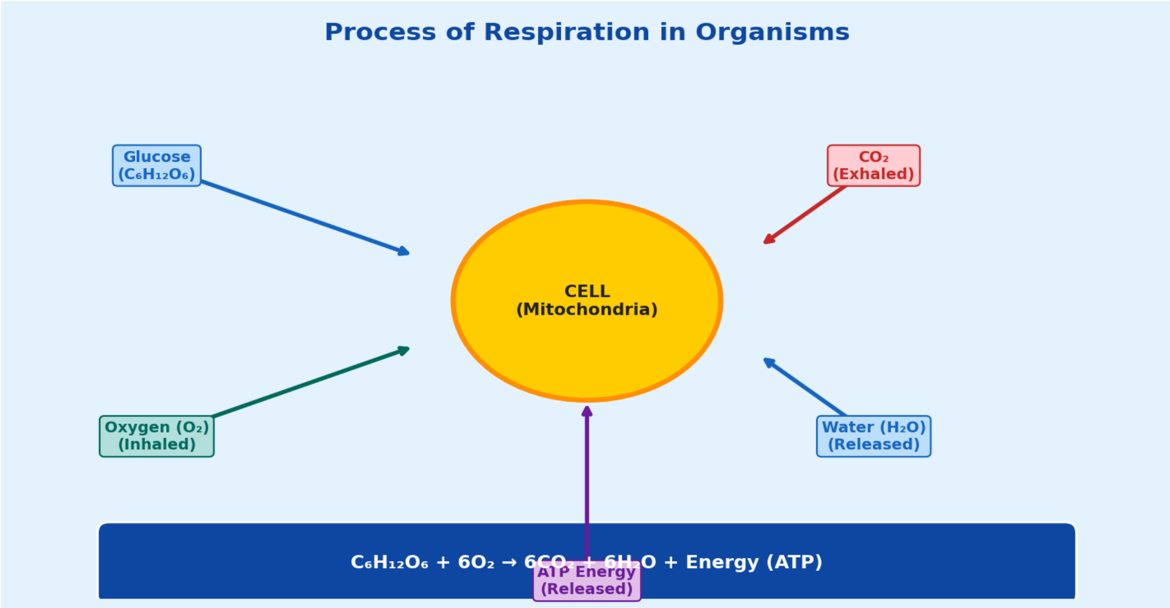

Fig 2.1 — Process of Cellular Respiration

2.2 Aerobic Respiration

Aerobic Respiration is the type of respiration that requires **oxygen**. It is the most efficient form of respiration, releasing the maximum amount of energy from glucose. It occurs in the **mitochondria** of cells.

C■H■■O■ + 6O■	■■■■■■■■■■■■■■■	6CO■ + 6H■O + 38 ATP

Steps of Aerobic Respiration

- **Step 1 — Glycolysis** (in cytoplasm): Glucose (6C) is split into 2 molecules of Pyruvate (3C). Produces 2 ATP.
- **Step 2 — Krebs Cycle** (in mitochondria): Pyruvate is fully broken down into CO■. More ATP and electron carriers made.
- **Step 3 — Electron Transport Chain** (inner mitochondrial membrane): Electrons generate a large amount of ATP. Most of the 38 ATP is produced here.

Why is aerobic respiration more efficient?

Because glucose is completely oxidised into CO■ and water, releasing ALL of its stored chemical energy. The complete breakdown yields 38 ATP molecules per glucose molecule, compared to only 2 ATP in anaerobic respiration.

2.3 Anaerobic Respiration

Anaerobic Respiration occurs in the **absence of oxygen**. It is also called {B("fermentation")} in microorganisms. Glucose is only partially broken down, so less energy is released.

Two Types of Anaerobic Respiration

Type	Organisms	Products	ATP
Lactic Acid Fermentation	Animal muscle cells, some bacteria	Lactic acid + 2 ATP	2
Alcoholic Fermentation	Yeast, some fungi	Ethanol + CO■ + 2 ATP	2

Lactic Acid in Muscles

During vigorous exercise, muscles work very fast and oxygen supply becomes insufficient. Muscle cells switch to anaerobic respiration and produce lactic acid. Accumulation of lactic acid causes **muscle cramps and fatigue**. After rest, blood supply restores oxygen and lactic acid is broken down.

Fermentation by Yeast

- Yeast performs anaerobic respiration producing ethanol (alcohol) and CO■.
- Equation: C■H■■O■ → 2C■H■OH + 2CO■ + Energy
- CO■ makes bread rise (dough expands) — baking uses this.
- Ethanol is the basis of wine, beer, and other alcoholic drinks — fermentation industry.
- Fermentation is also used to make curd, vinegar, and cheese.

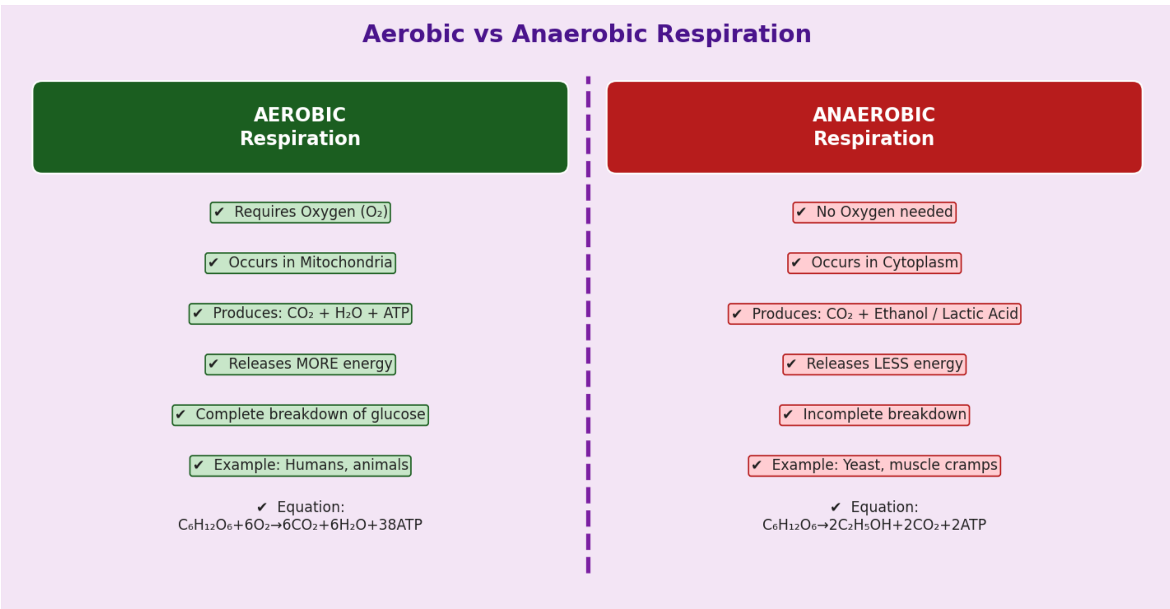

Fig 2.2 — Aerobic vs Anaerobic Respiration Compared

2.4 Breathing in Different Organisms

Different organisms have evolved different organs for breathing — to bring oxygen to their cells and remove carbon dioxide. The method depends on the organism's body structure and environment.

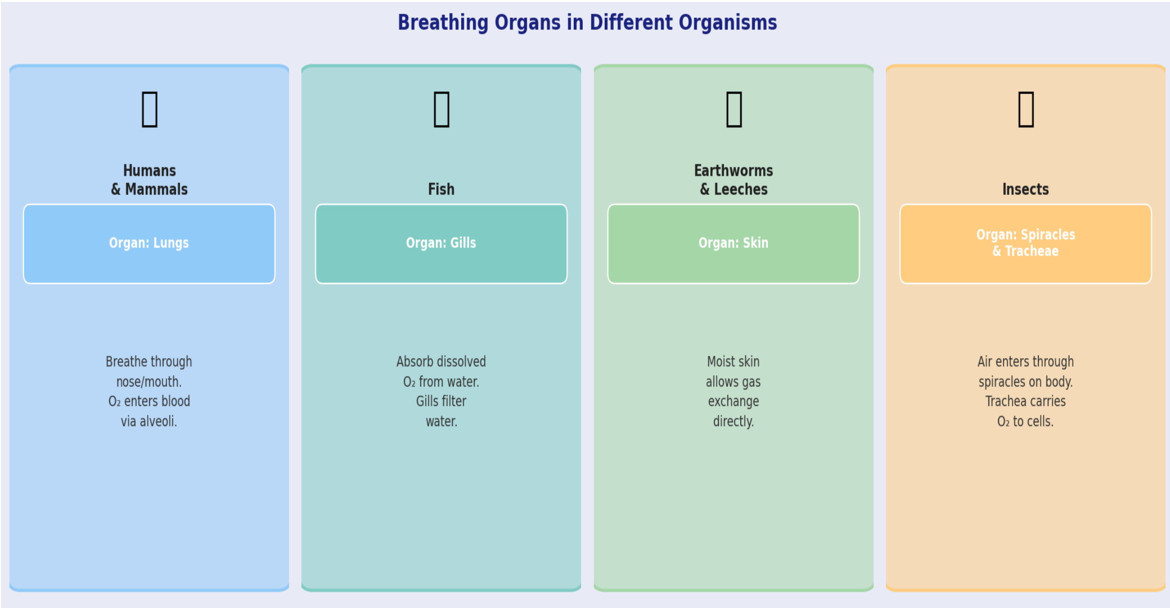

Fig 2.3 — Breathing Organs in Different Organisms

1. Humans and Mammals — Lungs

- Air enters through nose/mouth → trachea → bronchi → lungs.
- Gas exchange occurs in tiny air sacs called **alveoli** in the lungs.
- O■ passes into blood; CO■ passes from blood into alveoli → exhaled.
- Ribs and diaphragm create inhalation/exhalation movements.

2. Fish — Gills

- Fish do not breathe air directly — they use dissolved oxygen in water.
- Water enters through the mouth → passes over **gills** → exits through gill slits.
- Gills have rich blood supply; O■ from water diffuses into blood; CO■ moves out.

3. Earthworms — Skin (Cutaneous Respiration)

- Earthworms have no special respiratory organ.
- They breathe through their **moist skin** — gases diffuse in and out.
- The skin must remain moist for this to work — hence earthworms die in dry conditions.

4. Insects — Spiracles and Tracheae

- Insects have tiny holes on the body surface called **spiracles**.
- Air enters spiracles → carried by a network of tubes called **tracheae** to all cells.
- Tracheal system delivers O■ directly to cells — no blood carrier involved.

5. Plants — Stomata

- Plants also respire — in all cells, day and night.
- Gas exchange occurs through stomata in leaves and lenticels in stems.
- During night: only respiration occurs (takes O■, gives CO■).
- During day: photosynthesis rate > respiration rate, so net O■ is released.

2.5 Human Respiratory System in Detail

Nostrils	Air enters here. Hairs and mucus filter dust and microbes.
Nasal Cavity	Warms and humidifies the incoming air.
Pharynx	Common passage for food and air.
Larynx (Voice Box)	Helps in producing sound. Contains vocal cords.
Trachea (Windpipe)	Carries air from larynx to bronchi. Held open by C-shaped cartilage rings.
Bronchi	Trachea divides into left and right bronchus, entering each lung.
Bronchioles	Bronchi branch into smaller bronchioles inside the lung.
Alveoli	Tiny, thin-walled air sacs at the end of bronchioles. Site of gas exchange. Very large surface area (about 70 m^2).
Diaphragm	Dome-shaped muscle below lungs. Contracts during inhalation (flattens), relaxes during exhalation (moves up).

Mechanism of Breathing

Inhalation (Breathing in):

- Diaphragm contracts (moves down) and ribs move upward and outward.
- Volume of chest cavity increases → air pressure inside decreases.
- Air rushes in from outside to equalise pressure.

Exhalation (Breathing out):

- Diaphragm relaxes (moves up) and ribs move downward and inward.

- Volume of chest cavity decreases → air pressure increases.
- Air is pushed out.

■ *Normal breathing rate in humans at rest: 15–18 breaths per minute. During exercise it increases to 25–35 breaths per minute. A person breathes about 15,000–20,000 times every day!*

2.6 Yeast, Fermentation & Applications

Yeast is a single-celled fungus. It can survive both in the presence and absence of oxygen. Under anaerobic conditions (no oxygen), it carries out alcoholic fermentation.

Applications of Fermentation

Bread Making	Yeast added to dough. CO■ produced causes dough to rise → soft, spongy bread.
Alcohol Production	Fermentation of sugars (grapes, barley) → ethanol → wine, beer.
Curd / Yoghurt	Lactobacillus bacteria ferment milk → lactic acid → curd.
Vinegar	Acetic acid bacteria ferment ethanol → acetic acid (vinegar).
Cheese	Different microbes ferment milk proteins and fats → various cheeses.
Biofuel	Fermentation of biomass → bioethanol, used as fuel.

2.7 Photosynthesis vs Respiration

Photosynthesis vs Respiration — Quick Comparison

Feature	Photosynthesis	Respiration
Occurs in	Only green plants	All living organisms
Raw materials	CO_2 + H_2O + Sunlight	Glucose + O_2
Products	Glucose + O_2	CO_2 + H_2O + Energy
Energy	Stored (anabolic)	Released (catabolic)
Site	Chloroplast	Mitochondria
Time	Daytime only	Day and Night (always)
Gas exchange	Takes CO_2, releases O_2	Takes O_2, releases CO_2
Net result	Food synthesis	Energy release (ATP)

Fig 2.4 — Comparison: Photosynthesis vs Respiration

2.8 Key Definitions — Chapter 2

Respiration	The biochemical process of breaking down glucose to release energy (ATP) in all living cells.
Aerobic	Requires oxygen; produces CO■, H■O, and 38 ATP from one glucose.
Anaerobic	Does not require oxygen; produces lactic acid or ethanol + CO■, and only 2 ATP.
ATP	Adenosine Triphosphate — the energy currency of cells. Stores and supplies energy.
Mitochondria	Organelle in cells where aerobic respiration occurs. Called "Powerhouse of Cell."
Alveoli	Tiny air sacs in human lungs where gas exchange (O■ ↔ CO■) takes place.
Spiracles	Small holes on the body of insects for gas exchange.
Tracheae	Network of air tubes in insects that deliver O■ directly to cells.
Gills	Respiratory organs of fish; absorb dissolved oxygen from water.
Fermentation	Anaerobic breakdown of glucose by yeast to produce ethanol and CO■.
Lactic acid	Produced in muscle cells during intense exercise via anaerobic respiration; causes cramps.

Glycolysis	First step of respiration: glucose → pyruvate, occurs in cytoplasm.
Diaphragm	Dome-shaped muscle below the lungs that helps in breathing.

2.9 Important Questions & Answers

Q	Answer
What is respiration? How is it different from breathing?	Respiration is the biochemical breakdown of glucose to release energy (ATP) in cells. Breathing is the physical intake of air. Breathing is just one part of the overall respiratory process.
Write the equation for aerobic respiration.	C■H■■O■ + 6O■ → 6CO■ + 6H■O + 38 ATP (Energy). It occurs in mitochondria and requires oxygen.
What is anaerobic respiration? Give examples.	Anaerobic respiration occurs without oxygen. Examples: (1) Yeast: Glucose → Ethanol + CO■ + 2ATP. (2) Muscle cells: Glucose → Lactic Acid + 2ATP.
Why do muscles cramp during exercise?	During intense exercise, O■ supply is insufficient. Muscles perform anaerobic respiration → lactic acid builds up → causes pain/cramps.
How do fish breathe?	Fish use gills. Water enters through the mouth and passes over gills. Dissolved O■ in water diffuses into the blood; CO■ diffuses out.
How do earthworms breathe?	Earthworms breathe through their moist skin (cutaneous respiration). Gases diffuse directly through the skin surface. The skin must stay moist.
How do insects breathe?	Insects breathe through spiracles — small holes on their body. Air enters spiracles and is carried to all cells through a network of tubes called tracheae.
What is the role of alveoli in human lungs?	Alveoli are tiny air sacs in the lungs where gas exchange occurs. O■ from inhaled air diffuses into blood capillaries; CO■ from blood diffuses into the alveoli and is exhaled.

What is fermentation? State its uses.	Fermentation is anaerobic respiration by yeast: Glucose → Ethanol + CO_2. Uses: bread making (CO_2 makes bread rise), alcohol production (wine, beer), making curd, vinegar, and cheese.
Why is mitochondria called the powerhouse of the cell?	Mitochondria is where aerobic respiration takes place and most of the ATP (energy) is produced. Since it generates energy for all cellular activities, it is called the powerhouse.
Compare aerobic and anaerobic respiration.	Aerobic: needs O_2, complete breakdown, products are CO_2+H_2O, 38 ATP, occurs in mitochondria. Anaerobic: no O_2, incomplete breakdown, products are lactic acid or ethanol+CO_2, 2 ATP, occurs in cytoplasm.

Quick Revision Summary

■ KEY POINTS

- ❖ CHAPTER 1 RECAP: Plants make food by photosynthesis using CO■ + H■O + sunlight → Glucose + O■.
- ❖ Chlorophyll (in chloroplasts) absorbs sunlight. Stomata allow gas exchange.
- ❖ Parasitic: Cuscuta; Insectivorous: Pitcher Plant, Venus Flytrap; Symbiotic: Lichen, Rhizobium.
- ❖ Saprotrophs (Fungi, Bacteria) decompose dead matter → recycle nutrients.
- ❖ Rhizobium in legume roots fixes atmospheric N■ → enriches soil.

■ KEY POINTS

- ❖ CHAPTER 2 RECAP: Respiration breaks down glucose → energy (ATP). Occurs in all cells 24/7.
- ❖ Aerobic: needs O■ → 38 ATP + CO■ + H■O. Anaerobic: no O■ → 2 ATP + lactic acid or ethanol.
- ❖ Humans: lungs; Fish: gills; Earthworms: skin; Insects: spiracles + tracheae.
- ❖ Alveoli in lungs = site of gas exchange in humans.
- ❖ Yeast fermentation → bread, alcohol, curd. Muscle cramps = lactic acid buildup.


Photosynthesis vs Respiration — Quick Comparison

Feature	Photosynthesis	Respiration
Occurs in	Only green plants	All living organisms
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Net result	Food synthesis	Energy release (ATP)

Final Summary: Photosynthesis vs Respiration

■ All the Best for Your Exam! Revise key points daily and practise diagrams. ■